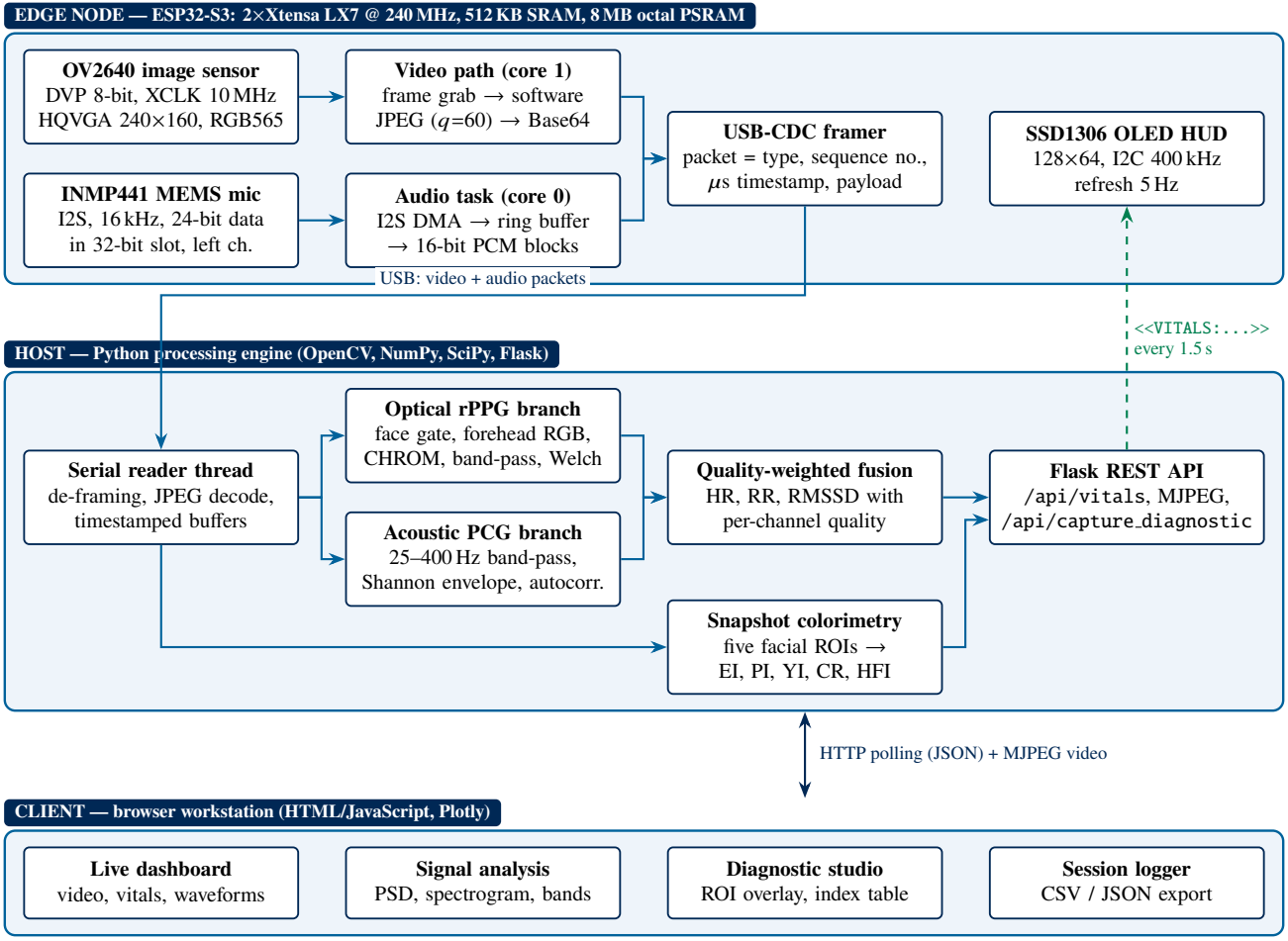

EDGE NODE — ESP32-S3: 2×Xtensa LX7 @ 240 MHz, 512 KB SRAM, 8 MB octal PSRAM
OV2640 image sensor
DVP 8-bit, XCLK 10 MHz
HQVGA 240×160, RGB565
Video path (core 1)
frame grab → software
JPEG (q=60) → Base64
INMP441 MEMS mic
I2S, 16 kHz, 24-bit data
in 32-bit slot, left ch.
Audio task (core 0)
I2S DMA → ring buffer
→ 16-bit PCM blocks
USB-CDC framer
packet = type, sequence no.,
μs timestamp, payload
SSD1306 OLED HUD
128×64, I2C 400 kHz
refresh 5 Hz
USB: video + audio packets
<<VITALS:...>>
every 1.5 s
HOST — Python processing engine (OpenCV, NumPy, SciPy, Flask)
Serial reader thread
de-framing, JPEG decode,
timestamped buffers
Optical rPPG branch
face gate, forehead RGB,
CHROM, band-pass, Welch
Acoustic PCG branch
25–400 Hz band-pass,
Shannon envelope, autocorr.
Quality-weighted fusion
HR, RR, RMSSD with
per-channel quality
Flask REST API
/api/vitals, MJPEG,
/api/capture_diagnostic
Snapshot colorimetry
five facial ROIs →
EI, PI, YI, CR, HFI
HTTP polling (JSON) + MJPEG video
CLIENT — browser workstation (HTML/JavaScript, Plotly)
Live dashboard
video, vitals, waveforms
Signal analysis
PSD, spectrogram, bands
Diagnostic studio
ROI overlay, index table
Session logger
CSV / JSON export